

664 Final Exam

Alex Caramanico

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1

What are the three stages of multiple imputation analysis? Please briefly describe what each stage does.

Answer: Multiple imputation involves three stages - 1) Imputation, 2) Analysis, and 3) Combining. 1) is the stage where the missing values of a data set are actually imputed, by using the posterior predictive distribution $f(Y_{miss}|Y_{obs})$. This step is repeated M times, hence “multiple” in the name of the approach. 2) consists of the analysis of *each* completed data set (the missing values replaced with imputed values). This of course gives M sets of results, and can include point estimates, standard error estimates, test statistics, etc. 3) is where the M sets of results are pooled together, so that a single set can be used for inference. This is usually done with Rubin’s combining rules, and allows for convenient analysis of the data with strong statistical backing due to the variation in the M imputed data sets. The theoretical justification to this is that the posterior distribution of some estimand can be approximated using averages, since said posterior is by definition a difficult-to-compute integral. (In our case, it would be impossible to compute because there are missing values).

2

What are the Rubin’s combining rules for multiple imputation analysis. Please use either formula or English to describe the essence for the rules.

Answer: The formulas for Rubin’s combining rules are as follows, with Q representing some estimand of interest:

$$a) \quad \bar{Q}_M = \frac{1}{M} \sum_{m=1}^M \hat{Q}_m$$

$$b) \quad T_M = \bar{U}_M + (1 + \frac{1}{M})B_M$$

a) represents the “average” of the estimates, which is a simple arithmetic mean. $\bar{Q}_M$ denotes this final estimate, while $\hat{Q}_m$ represents the m th estimate from the m th imputed data set. Of course, M is the total number of multiple imputations that have been made.

b) This metric is the total variance of the combined estimate, which is a weighted sum of the within-imputation variance $\bar{U}_M$ and between-imputation variance B_M . The former is taken as the average variance of the point estimates across each of the imputations. It is extremely important to consider *both* sources of variation when using multiple imputation.

3

We have learned one major R library for conducting multiple imputation analysis. What is the name of the library (10 pt)?

Answer: MICE

4

Please describe the basic idea behind predictive mean matching imputation. What are the advantages?

Answer: Predictive mean matching (PMM) is a very popular robust multiple imputation procedure which guides how imputed values are sampled. It closely resembles the “nearest-neighbor” idea, where a prediction is made for some unknown output based off known outputs from observations “near” it. PMM makes use of predictions from posterior distributions, then samples from *observed* values whose predicted value is close to the incomplete case. For each incomplete case, PMM randomly draws an observation from a subset of observed cases that have predicted values close to that of the incomplete case. Usually, the predicted values come from parameter estimates based on complete-case analysis. The value of the item (which corresponds to the missing value) from the selected observed case replaces the missing value. In other words, missing values are imputed by what is deemed to be similar in the other variables.

The approach is advantageous because it is much more flexible than hot deck imputation (despite their similarities), allowing for appropriate (and controlled) model set-ups for the posterior draws. Additionally, it is semi-parametric and therefore robust to model mis-specifications, preserves original data scales in imputation (only observed values are used), and is easily available in many software packages. It should be noted as well that PMM is preferred in situations where the assumption of normality (as is required for FCS) is violated.

5

Please describe the basic idea of fully conditional specification procedures for multivariate multiple imputation. What are the advantages?

Answer: Fully conditional specification (FCS) is an alternate method for posterior draws in multiple imputation. Essentially, the procedure uses all information we actually have to alternatively impute missing values. That is, if we have missing values for X and Y , we can impute Y with some model using initial guesses (means are usually sufficient) for X . Then, using the now “completed” Y , we can switch the order and predict the once-missing values for X . Each variable is “momentarily” completed so that it can be used to impute the other(s), and this process repeats until convergence. This can be extended to several variables in a manner very reminiscent of the Gibbs Sampling algorithm.

There are numerous strengths for FCS as well, the strongest perhaps being its flexibility for univariate *and* multivariate missing data problems. It also gives the analyst a great deal of flexibility as each variable can be imputed with a different modeling approach (such as multivariate normal or PMM). This provides a great deal of control and allows for sensitivity analyses, a very important consideration in missing data analysis.

6

Suppose a dataset has around 20% missing cases, you want to conduct a multiple imputation analysis. What is the number of imputations you want to do? State the rationale behind your decision.

Answer: You would want to do around $M = 20$ imputations, as the general rule of thumb for multiple imputation is to do $M = 100 \times p_{miss}$, where p_{miss} is the proportion of missingness in the data. Typically, more imputations are better, as it is expected for estimates to converge to true values as more variance is captured due to repeated sampling. However, too many imputations could be computationally prohibitive if data sets are extremely large (with observations and/or variables) or if there is a very large amount missing. Thus, the aforementioned rule of thumb is commonly used as a compromise between these two points of emphasis.

7

Please briefly state what you have learned from this missing data analysis course that you didn't know before. And what topic areas in missing data analysis that you wish to learn more but are not able to get much in the class.

Answer: Before this class, I knew nothing about multiple imputation, even coming from a statistics background. Specifically, I had not heard of predictive mean matching or fully conditional specification procedures, at least as it relates to handling missing data. I had learned about full conditionals in the context of Gibbs Sampling for complex distributions, though. I also learned about the various missing data mechanisms (MCAR, MAR, MNAR), which will certainly be patterns I test for in the future. Thirdly, I learned how likelihood methods (which I am familiar with from statistics) can be applied in missing data scenarios.

As far as additional topic areas, we discussed briefly the applications of Bayesian methods to missing data analysis. This is a specific interest of mine, and the idea of using sensitivity analysis in an MNAR situation seems practical and intriguing (though potentially more advanced). I am also curious about the potential for hierarchical Bayesian modeling for missing data, though I understand this could be “overkill” and is certainly outside the scope of the course. On a more general note, it may have been nice to see a wider range of data sets. Perhaps this could have been done with the homework assignments, where various data types (with several missing data patterns) were provided for analysis. Though I chose the Air Quality data set myself for most of the assignments, I wish I had applied the methods to several data sets (or at least more interesting ones).

Overall, this course will be extremely useful to me because I have run into *many* data sets with missing values and was always skeptical about removing them or imputing the mean. Both of these were revealed to have extremely problematic effects, so I will most likely avoid them and stick to multiple imputation in the future. Fully conditional specification appears to be one of the more flexible and appropriate methods for many cases, so I am sure I will use it. I look forward to applying the techniques I learned in this class to my future work across all kinds of data - I believe it will make me a better statistician.

8

Suppose you are working with an investigator who is interested in using RANDS dataset to analyze the included variables. More information about the dataset and the meaning of these variables can be found in Lecture 12.

Note: I talked about this example using the survey procedure in my class on 04/28. Considering that some of you might not be very familiar with the survey statistics procedure, you can freely choose to use either survey procedure or non-survey procedure to impute and analyze these data. In non-survey procedure, you can ignore the survey weight and survey strata when imputing and analyzing the data. Either way is fine, but please state your option clearly in your report.

This investigator noticed that both income variable (INCOME) and marital status (MARITAL_NEW) have missing values. They are interested in the following analyses:

Load Libraries

```
library(mice)
library(tidyverse)
library(norm)
library(survey)
```

```
library(mitools)
library(plotrix)
```

a

How much missing data do the two variables have? Do other variables in the dataset also have missing values?

```
rand_data <- read.csv("C:/Users/jacma/OneDrive/school work/UMD/EPIB 664 Missing Data/rands_mi_example.csv")
rand_data$EDUC <- as.factor(rand_data$EDUC) # change to categorical variables
rand_data$GENDER <- as.factor(rand_data$GENDER)
rand_data$INTERNET <- as.factor(rand_data$INTERNET)
rand_data$MARITAL_NEW <- as.factor(rand_data$MARITAL_NEW)
rand_data$S_VSTRAT_COMBINE <- as.factor(rand_data$S_VSTRAT_COMBINE)
n_missing <- rand_data %>% apply(2,function(col){sum(is.na(col))})
data.frame(`Num Missing` = n_missing,
           Proportion = n_missing / nrow(rand_data))
```

##		Num.Missing	Proportion
##	AGE	0	0.0000000
##	EDUC	0	0.0000000
##	GENDER	0	0.0000000
##	HHSIZE	0	0.0000000
##	INCOME	1201	0.2200440
##	INTERNET	0	0.0000000
##	WEIGHT_CALIBRATED	0	0.0000000
##	S_VSTRAT_COMBINE	0	0.0000000
##	MARITAL_NEW	1252	0.2293881

Income and marital status have 1201 and 1252 missing values, constituting 22% and 22.9% of the total observations in the data, respectively. Every other variable is completely observed with no missing values.

b

The investigator is interested in estimating the mean of INCOME and the proportion of MARITAL_NEW. Please perform the multiple imputation analysis for him/her. And please also compare the multiple imputation analysis results with complete-case analysis results.

```
# Multiple Imputation
n_imputations <- max(n_missing/nrow(rand_data)*100) %>% round()
relevant_info <- rand_data[,-c(7,8)] # weighted info and strata not needed
mean_imp1 <- mice(relevant_info,
                 method=c("", "", "", "", "pmm", "", "logreg"),
                 maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~1)) %>%
  pool() %>%
  summary()
```

```

mean_imp2 <- mice(relevant_info,
  method=c("", "", "", "", "pmm", "", "pmm"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~1)) %>%
  pool() %>%
  summary()

mean_imp3 <- mice(relevant_info,
  method=c("", "", "", "", "norm", "", "logreg"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~1)) %>%
  pool() %>%
  summary()

mean_imp4 <- mice(relevant_info,
  method=c("", "", "", "", "norm", "", "pmm"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~1)) %>%
  pool() %>%
  summary()

# Comparisons
cbind(Complete_Case = c(Mean=rand_data$INCOME %>% mean(na.rm=T),
  StdErr=rand_data$INCOME %>% std.error(na.rm=T)),
  PMM_LogReg_MI = c(mean_imp1$estimate,
    mean_imp1$std.error),
  PMM_PMM_MI = c(mean_imp2$estimate,
    mean_imp2$std.error),
  Norm_LogReg_MI = c(mean_imp3$estimate,
    mean_imp3$std.error),
  Norm_PMM_MI = c(mean_imp4$estimate,
    mean_imp4$std.error))

```

Non-Survey Procedure

	Complete_Case	PMM_LogReg_MI	PMM_PMM_MI	Norm_LogReg_MI	Norm_PMM_MI
## Mean	9.96499883	10.1993643	10.2015709	10.1783200	10.1838525
## StdErr	0.06531939	0.0679681	0.0664739	0.0653206	0.0668306

```
rm(mean_imp1, mean_imp2, mean_imp3, mean_imp4)
```

The first table of results are from the non-survey approach to the imputation problem. The columns of the results are named by the methods that were used under fully conditional specification - the first method in the column name being applied to INCOME, the second being applied to MARITAL_NEW. Logistic regression seems the most natural choice for the latter (since it is binary), but PMM or Norm could be reasonable for INCOME. In any case, all results are extremely similar across the FCS imputations - the largest difference in mean estimate is only about 0.023 between PMM_PMM and Norm_LogReg, and the standard errors are extremely similar (and small relative to the mean estimates) across every approach. However, all of the FCS fits have a larger mean estimate than complete case analysis, which could be strong evidence that removing all NAs leads to an underestimate of mean income. The CC mean is outside of three standard errors from the next-closed mean estimate. The small standard error for Norm_LogReg, along

with its intuitive justification, might make it the most appealing approach for estimating the mean of income.

```
# Complete-Case
rands_design <- svydesign(id=~1, strata=~S_VSTRAT_COMBINE,
                        weights=~WEIGHT_CALIBRATED, data=rand_data)

cc_survey_mean <- svymean(~INCOME, na.rm=TRUE, design=rands_design,
                        df=sum(!is.na(rand_data$INCOME))-
                          rand_data$S_VSTRAT_COMBINE %>%
                            unique() %>%
                              length())

# Multiple Imputation
income_impute <- mice(rand_data, method=c("", "", "", "", "norm", "", "", "", "logreg"),
                     maxit=5, m=n_imputations, seed=875, print=F)

income_impute_data <- complete(income_impute, action='long')

all_imp <- imputationList(lapply(1:n_imputations, function(n) {
  subset(income_impute_data, subset=.imp==n)}))

rands_impute_whole_design <- svydesign(id=~1, strata=~S_VSTRAT_COMBINE,
                                     weights=~WEIGHT_CALIBRATED, data=all_imp)

mi_survey_mean <- with(rands_impute_whole_design, svyglm(INCOME~1)) %>%
  MIcombine(print=F)

rbind(Complete_Case=c(cc_survey_mean,
                      StdErr=attributes(cc_survey_mean)$var %>% sqrt()),
      Norm_LogReg_MI = c(mi_survey_mean$coefficients,
                        StdErr=mi_survey_mean$variance %>% sqrt()))
```

Survey Procedure

```
##              INCOME      StdErr
## Complete_Case 10.17138 0.09836198
## Norm_LogReg_MI 10.40571 0.09020503

rm(cc_survey_mean, income_impute, income_impute_data, all_imp, mi_survey_mean)
```

The complete-case result with the survey approach is very similar to the FCS non-survey results, both of which are considerably greater than the initial complete-case estimate. The multiple imputation result continues the pattern seen in the non-survey analysis, where the estimate is increased after imputing the missing data. The standard error is also notably lower for the MI model, which may suggest that MI is more appropriate for this problem.

c

The investigator is also interested in running a regression analysis of INCOME on other predictors (i.e., age, education, gender, internet access, marital status, and household size). Please perform multiple imputation

analysis for him/her. And please also compare the multiple imputation analysis results with complete-case results.

```
# Complete-case analysis
cc_model <- lm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW,data=na.omit(rand_data)) %>% summary()

# Multiple Imputation
reg_imp1 <- mice(rand_data,
  method=c("", "", "", "", "pmm", "", "", "", "logreg"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW)) %>%
  pool() %>%
  summary()

reg_imp2 <- mice(rand_data,
  method=c("", "", "", "", "pmm", "", "", "", "pmm"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW)) %>%
  pool() %>%
  summary()

reg_imp3 <- mice(rand_data,
  method=c("", "", "", "", "norm", "", "", "", "logreg"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW)) %>%
  pool() %>%
  summary()

reg_imp4 <- mice(rand_data,
  method=c("", "", "", "", "norm", "", "", "", "pmm"),
  maxit=5, m = n_imputations, seed=875, print=F) %>%
  with(exp=lm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW)) %>%
  pool() %>%
  summary()

# Comparisons
## Parameter Estimates
cbind(Complete_Case=cc_model$coefficients[,1],
  PMM_LogReg_MI=reg_imp1$estimate,
  PMM_PMM_MI=reg_imp2$estimate,
  Norm_LogReg_MI=reg_imp3$estimate,
  Norm_PMM_MI=reg_imp4$estimate)
```

Non-Survey Procedure

##	Complete_Case	PMM_LogReg_MI	PMM_PMM_MI	Norm_LogReg_MI
## (Intercept)	4.591699225	4.494907250	4.462957024	4.477570793
## AGE	0.001085985	0.006427234	0.006335866	0.006194201
## EDUC3	1.179998088	1.290561641	1.299666379	1.305760503
## EDUC4	3.281442931	3.395897349	3.367942810	3.402168447

```
## GENDER2      -0.436971139  -0.411894916  -0.409704183  -0.388552412
## HHSIZE       0.265961386   0.138520550   0.155574118   0.136274143
## INTERNET1    2.490735084   2.441454986   2.456844717   2.439624051
## MARITAL_NEW1 2.675798598   2.740670838   2.703409683   2.754363175
##              Norm_PMM_MI
## (Intercept)  4.436988823
## AGE          0.007204667
## EDUC3        1.284015619
## EDUC4        3.418004140
## GENDER2      -0.425053878
## HHSIZE       0.159519647
## INTERNET1    2.456251240
## MARITAL_NEW1 2.670922987
```

Standard Error Estimates

```
cbind(Complete_Case=cc_model$coefficients[,2],
      PMM_LogReg_MI=reg_imp1$std.error,
      PMM_PMM_MI=reg_imp2$std.error,
      Norm_LogReg_MI=reg_imp3$std.error,
      Norm_PMM_MI=reg_imp4$std.error)
```

##	Complete_Case	PMM_LogReg_MI	PMM_PMM_MI	Norm_LogReg_MI	Norm_PMM_MI
## (Intercept)	0.348904904	0.30837155	0.31526665	0.310552070	0.31378682
## AGE	0.004296445	0.00403474	0.00384362	0.003762845	0.00383242
## EDUC3	0.154178322	0.13788968	0.13978783	0.136702840	0.13559613
## EDUC4	0.173208157	0.14681221	0.16148366	0.160349894	0.14605488
## GENDER2	0.122299465	0.10606116	0.10623023	0.106152411	0.11645025
## HHSIZE	0.055308038	0.04481752	0.04098326	0.042677984	0.04378641
## INTERNET1	0.164101512	0.14930612	0.14624309	0.148642581	0.14353872
## MARITAL_NEW1	0.133639897	0.13629377	0.13248637	0.129101845	0.14309796

P-Values

```
cbind(Complete_Case=cc_model$coefficients[,4],
      PMM_LogReg_MI=reg_imp1$p.value,
      PMM_PMM_MI=reg_imp2$p.value,
      Norm_LogReg_MI=reg_imp3$p.value,
      Norm_PMM_MI=reg_imp4$p.value)
```

##	Complete_Case	PMM_LogReg_MI	PMM_PMM_MI	Norm_LogReg_MI
## (Intercept)	1.396116e-38	5.549648e-38	1.397973e-34	1.293282e-37
## AGE	8.004668e-01	1.128252e-01	1.003346e-01	1.004648e-01
## EDUC3	2.557755e-14	1.118370e-18	3.338226e-18	1.006603e-19
## EDUC4	4.967168e-76	5.645913e-85	2.336265e-52	2.474027e-56
## GENDER2	3.580625e-04	1.186782e-04	1.321811e-04	2.784663e-04
## HHSIZE	1.587498e-06	2.321454e-03	1.698024e-04	1.555851e-03
## INTERNET1	2.468564e-50	7.874413e-45	1.887277e-49	7.206978e-47
## MARITAL_NEW1	3.214696e-84	5.664545e-41	4.626951e-44	7.684170e-53
## Norm_PMM_MI				
## (Intercept)	1.086482e-35			
## AGE	6.097072e-02			
## EDUC3	1.221285e-19			
## EDUC4	1.830640e-92			
## GENDER2	3.386476e-04			


```
## HHSIZE      3.345653e-04
## INTERNET1   4.237357e-55
## MARITAL_NEW 1.998638e-34
```

```
rm(cc_model,reg_imp1,reg_imp2,reg_imp3,reg_imp4)
```

The first tables show the parameter estimates for each FCS approach and their respective standard errors. We see that there are really no considerable changes across any of the methods - each parameter estimate is very similar and the standard errors are close in value. This is reflected by the third table which shows the p-values, where every model has *every* variable significant in explaining the variance in INCOME aside from AGE. Notably, the significance of EDUC4 is much “lesser” under the PMM_PMM and Norm_LogReg models, with a p-value 30 times larger in magnitude (despite remaining extremely small). MARITAL_NEW, while *very* significant under each model, has a much smaller p-value under the complete case approach than any FCS model. This could suggest that complete-case overestimates the effect/significance of the variable toward INCOME. Also worth noting is that Complete-Case analysis generally has the largest standard error estimates for each parameter, but this is due to the “increased” sample size of the imputed data.

Based on these results, it can be concluded pretty strongly that all variables have a significant effect on INCOME *except* for age. Further, all have a positive relationship with INCOME aside from GENDER. That is, being more educated, having a bigger house, having internet access, and being married all significantly *increase* the expected income of the individual, whereas being female (a value of 1 for GENDER) decreases expected income. This final interpretation owes to the negative estimate for GENDER’s parameter. The *most* statistically significant variables are GENDER, INTERNET, and MARITAL_NEW.

```
cc_reg <- svyglm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW, design=rands_design) %>%
  summary()

survey_regression <- with(rands_impute_whole_design,
  svyglm(INCOME~AGE+EDUC+GENDER+HHSIZE+INTERNET+MARITAL_NEW)) %>%
  MIcombine(print=F)
# ^ gives covariance estimates between variables, all are very small, disregard

cbind(Complete_Case = cc_reg$coefficients[,1],
      CC_SE = cc_reg$coefficients[,2],
      Norm_LogReg_MI=survey_regression$coefficients,
      MI_SE = survey_regression$variance %>% diag() %>% sqrt())
```

Survey Procedure

##	Complete_Case	CC_SE	Norm_LogReg_MI	MI_SE
## (Intercept)	4.210035431	0.519646051	4.40621860	0.453086308
## AGE	0.009838424	0.006512867	0.01253869	0.005650724
## EDUC3	1.275504417	0.230645857	1.42363051	0.199526072
## EDUC4	3.385691440	0.225106691	3.48376373	0.221698871
## GENDER2	-0.377417199	0.177614122	-0.37926643	0.151808626
## HHSIZE	0.303118520	0.074110424	0.12431777	0.060075842
## INTERNET1	2.897244163	0.249339666	2.75812685	0.228050159
## MARITAL_NEW1	2.256247969	0.198447600	2.38697306	0.198067699

```
rm(list=ls())
```

The interpretations of every parameter remain the same for *both* of the survey- based approaches. There are some slight differences between the complete-case survey method and the FCS survey method. Most parameter estimates are very alike, but the estimate for HHSIZE decreased. However, the standard error estimates are all smaller for the multiply-imputed estimates, but this is to be expected because the sample size is larger. The results being extremely similar across *all* methods (Survey, Non-Survey, Complete-Case and MI within each) suggests that this analytical problem is somewhat robust to the missing data. That is, complete-case analysis *may* be sufficient for the regression problem, though imputation certainly has its justifications.